GAIRA V7 — Phase 06.5

Latent Spectral Geometry Audit

What biochemical organisation emerges from the frozen CSM representation, independent of the curated ontology?

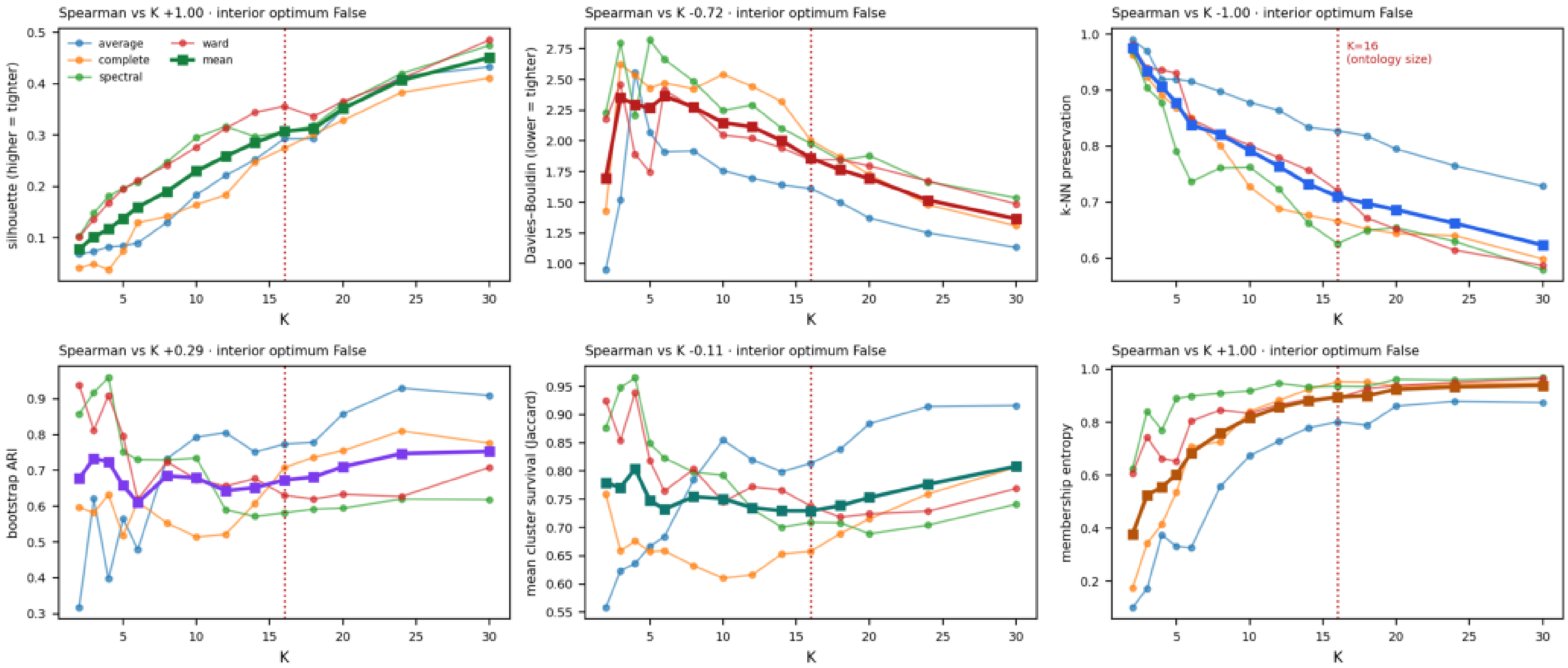
Status	COMPLETE — AUDIT ONLY, no architecture changed
Recommendation	Option A — current architecture retained
Unit of analysis	canonical molecule (154), not spectrum
Labels in construction	NONE — chemistry is an external validation target only
Preferred cluster count	NONE — 0 of 7 indices have an interior optimum
Chemistry vs acquisition	PERMANOVA R^2 0.452 vs excitation 0.118, source 0.040
Graph modularity	0.718 vs null 0.347 (z = 40)
Shape of the space	modular — hierarchy with continuous branch lengths
Coordinates vs hard ids	k-NN preservation 0.446 vs 0.237
Coordinate reproducibility	0.963
Coordinate robustness	mean cosine 0.958
Retrieval gain from coordinates	molecule Δ+0.016 (p = 0.180) · chemistry Δ+0.003 (p = 1.000) — NEITHER SIGNIFICANT
Agreement with the ontology	AMI 0.702 at K = 16 — but AMI is monotone in K and peaks at K = 24
Frozen inputs	LSM 208482d6… · CSM 0b4aa550… · engine 20d8bd99… — verified
Phase 07	NOT BEGUN
Completed	2026-08-07 02:45:45 UTC

Sources of record: reports/PHASE_06_5_LATENT_GEOMETRY_AUDIT.md and reports/PHASE_06_5_SCIENTIFIC_AUDIT.md
The Continuous Spectral Coordinates are retained as a SCIENTIFIC INSTRUMENT. They are not a chemistry probability, not a concentration, and they do not enter the GAIRA inference path.

Figures

Figure 01

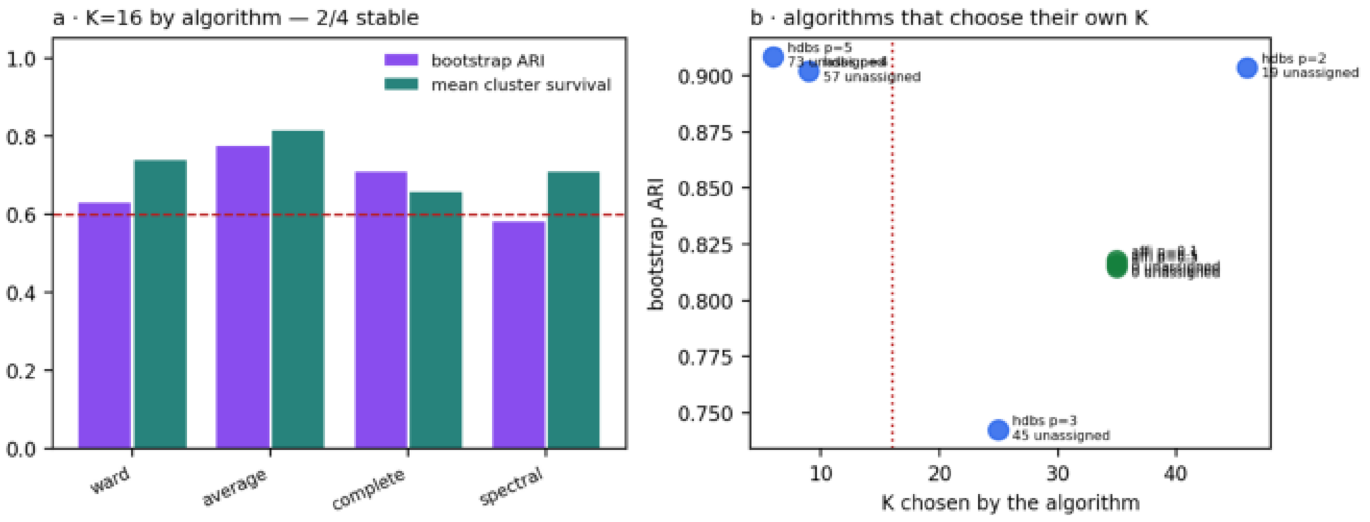
Figure 1 · Cluster stability across 14 K and 4 fixed-K algorithms — NO index has an interior optimum
every curve is monotone or flat in K: the geometry has no preferred cluster count, which is what a continuum looks like



Cluster stability across 14 values of K and four fixed-K algorithms. NO index has an interior optimum: silhouette rises monotonically to K=30 (Spearman +1.00), neighbour preservation falls monotonically (-1.00), membership entropy rises monotonically (+1.00). Bootstrap stability is U-shaped because coarse partitions and near-singleton partitions are both trivially reproducible. This is what a continuum looks like.

Figure 02

Figure 2 · Is K=16 genuinely stable, or merely convenient?



NO PREFERRED CLUSTER COUNT EXISTS

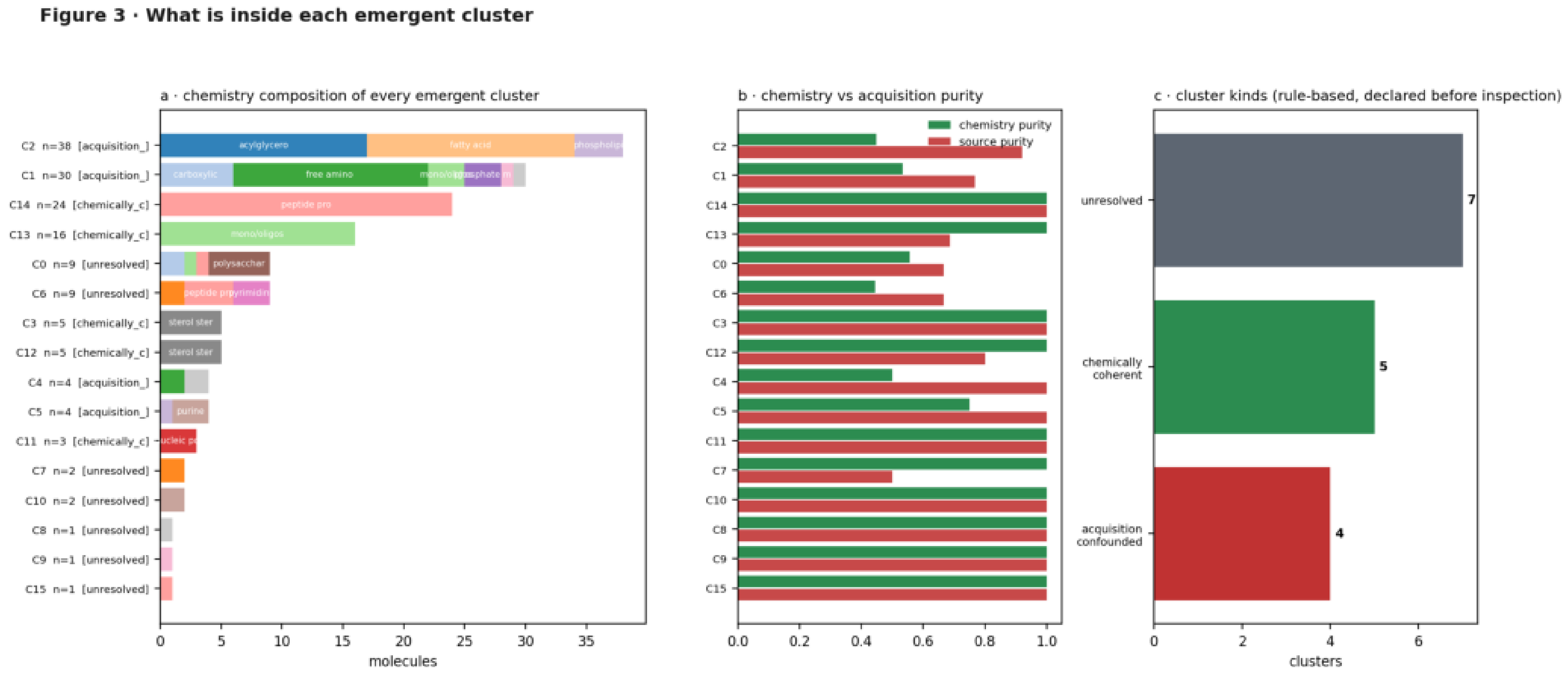
0 of 7 internal indices have an interior optimum.

Silhouette rises monotonically to K=30 (Spearman +1.00), neighbour preservation falls monotonically to K=30 (−1.00), membership entropy rises monotonically (+1.00). Every index is tracking granularity, not structure.

K=16 is therefore adopted as a REPORTING CONVENTION — it matches the curated ontology so Section 7's comparison is like-for-like — and NOT as a discovered optimum. An earlier rule that maximised bootstrap ARI over all K chose K=4, because a coarse partition is trivially reproducible: the same stability-without-informativeness trap that principle P-18 exists to catch.

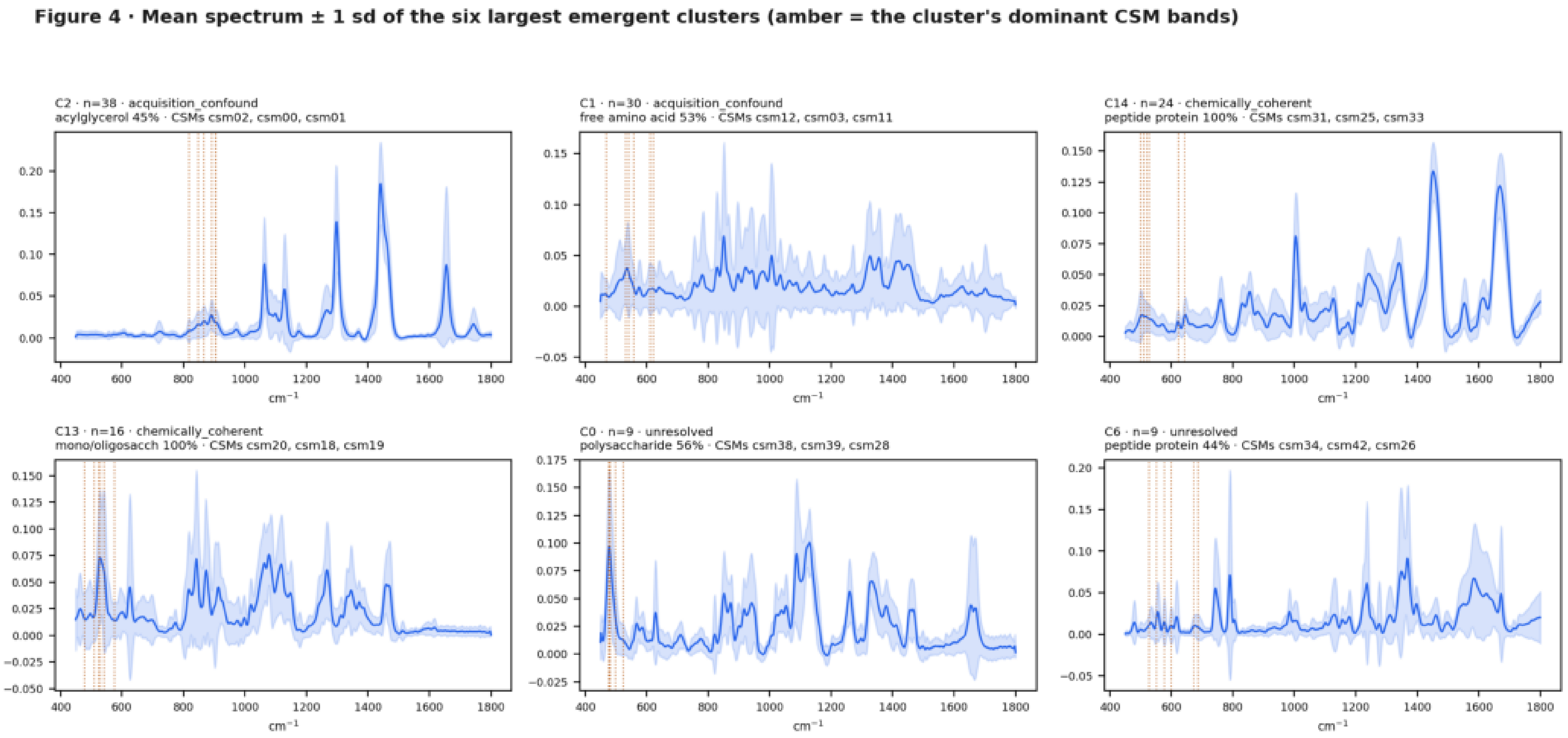
Is K=16 genuinely stable or merely convenient? Merely convenient: 2 of 4 algorithms call it stable and its mean bootstrap ARI (0.673) is unremarkable against its neighbours. The free-K algorithms choose 3-24, never 16. K=16 is adopted as a reporting convention for comparability with the ontology, not as a discovery.

Figure 03



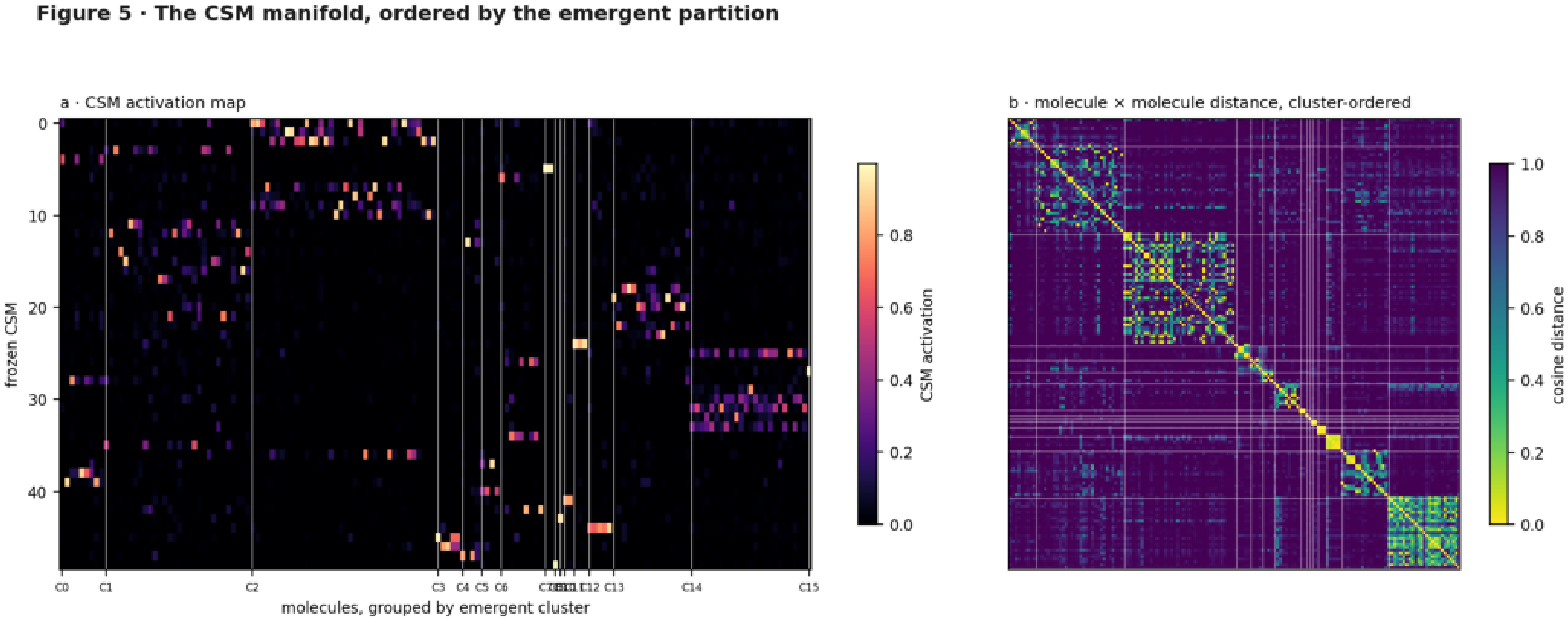
Composition of every emergent cluster, and the rule-based classification whose thresholds were declared before any cluster was inspected. Five clusters are 100% chemically pure without a label having been used; four are acquisition-confounded, their source or excitation purity exceeding their chemistry purity.

Figure 04



Mean spectrum +/- 1 sd of the six largest emergent clusters, with each cluster's dominant CSM bands marked. The clusters are spectroscopically coherent objects, not arbitrary partitions of a cloud.

Figure 05

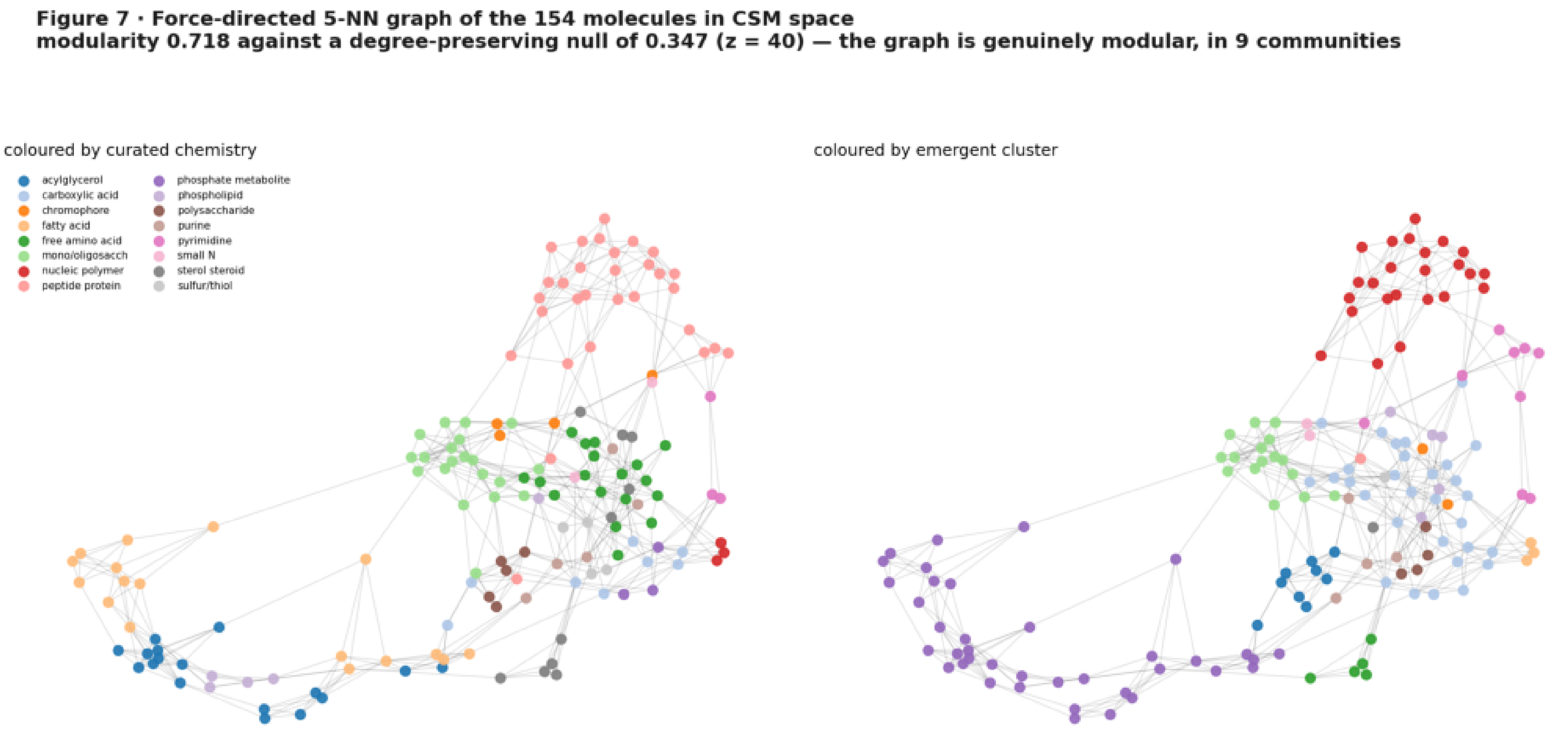


The CSM activation map and the molecule x molecule distance matrix, both ordered by the emergent partition. Block structure is visible and soft-edged.

Figure 06

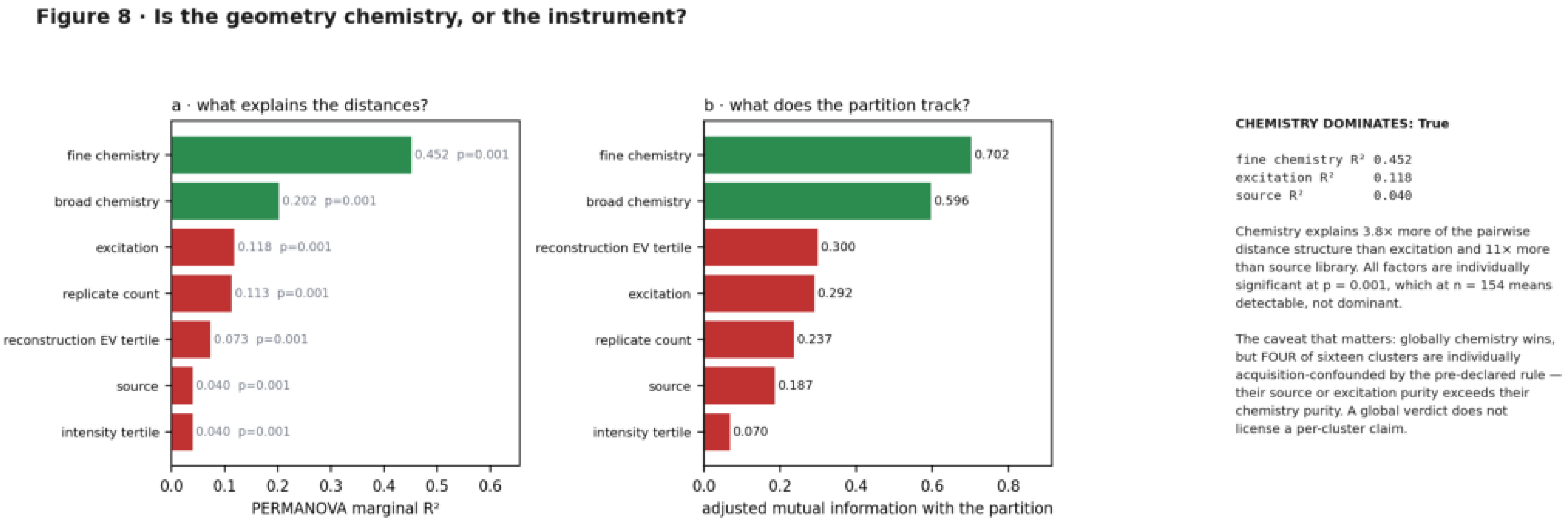
PCA, UMAP and MDS embeddings coloured by curated chemistry (top) and by emergent cluster (bottom). Visualisation only: PCA components 1-2 explain 15.5% of variance and UMAP distances must not be read quantitatively.

Figure 07



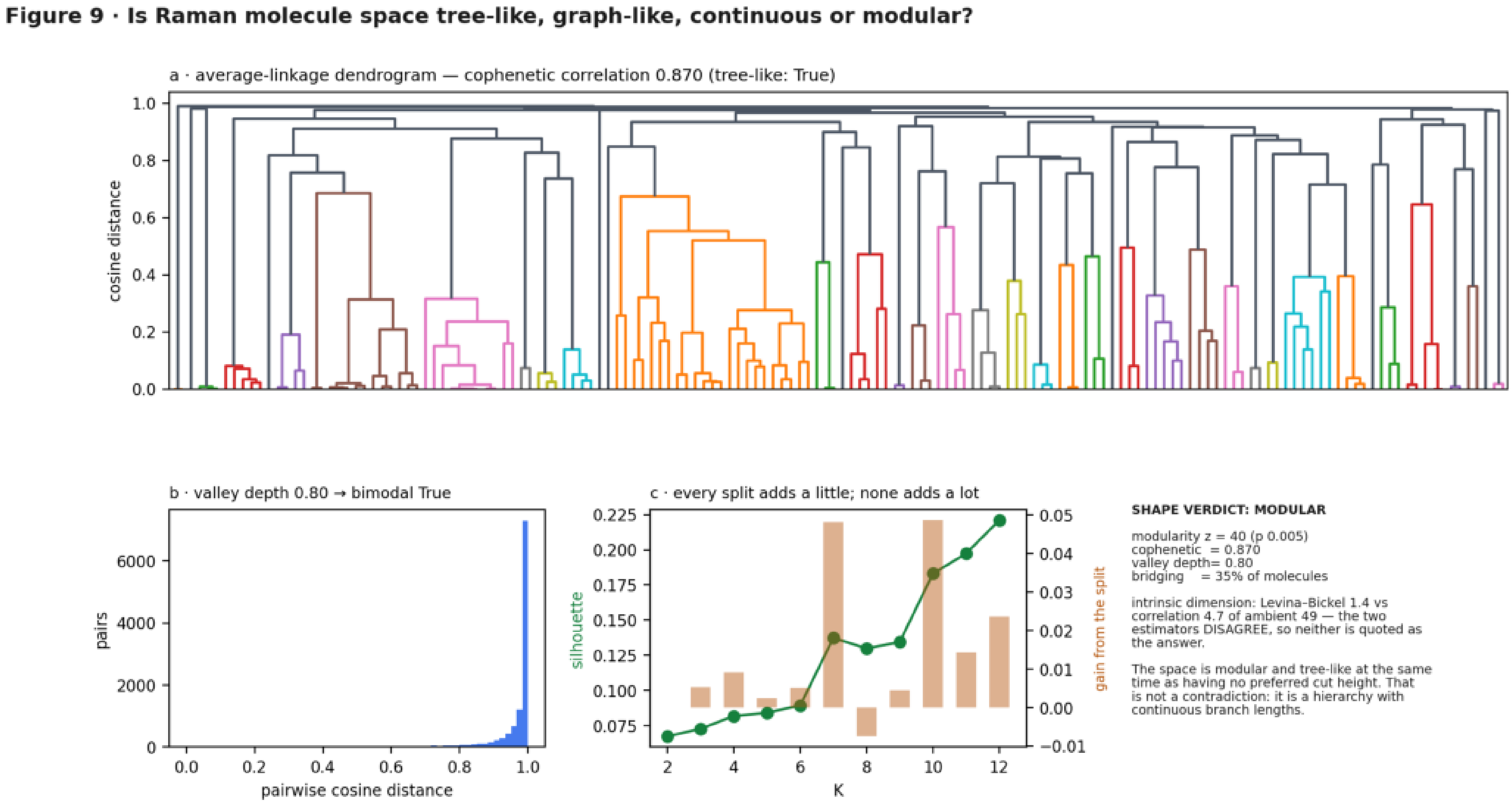
Force-directed 5-NN graph of the 154 molecules. Modularity 0.718 against a degree-preserving null of 0.347 (z = 40) - the graph is genuinely modular - while 35% of molecules have neighbours in more than one community.

Figure 08



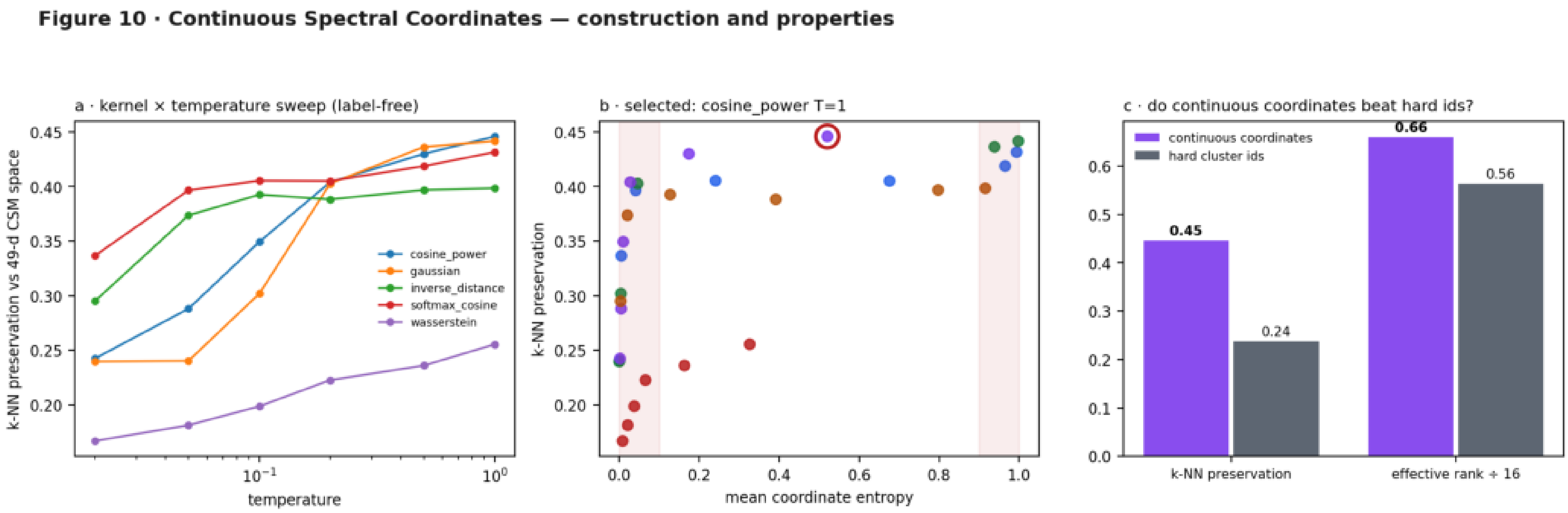
Is the geometry chemistry or the instrument? Chemistry explains 3.8x more of the distance structure than excitation and 11x more than source library. The caveat: globally chemistry wins, but four individual clusters are acquisition-confounded.

Figure 09



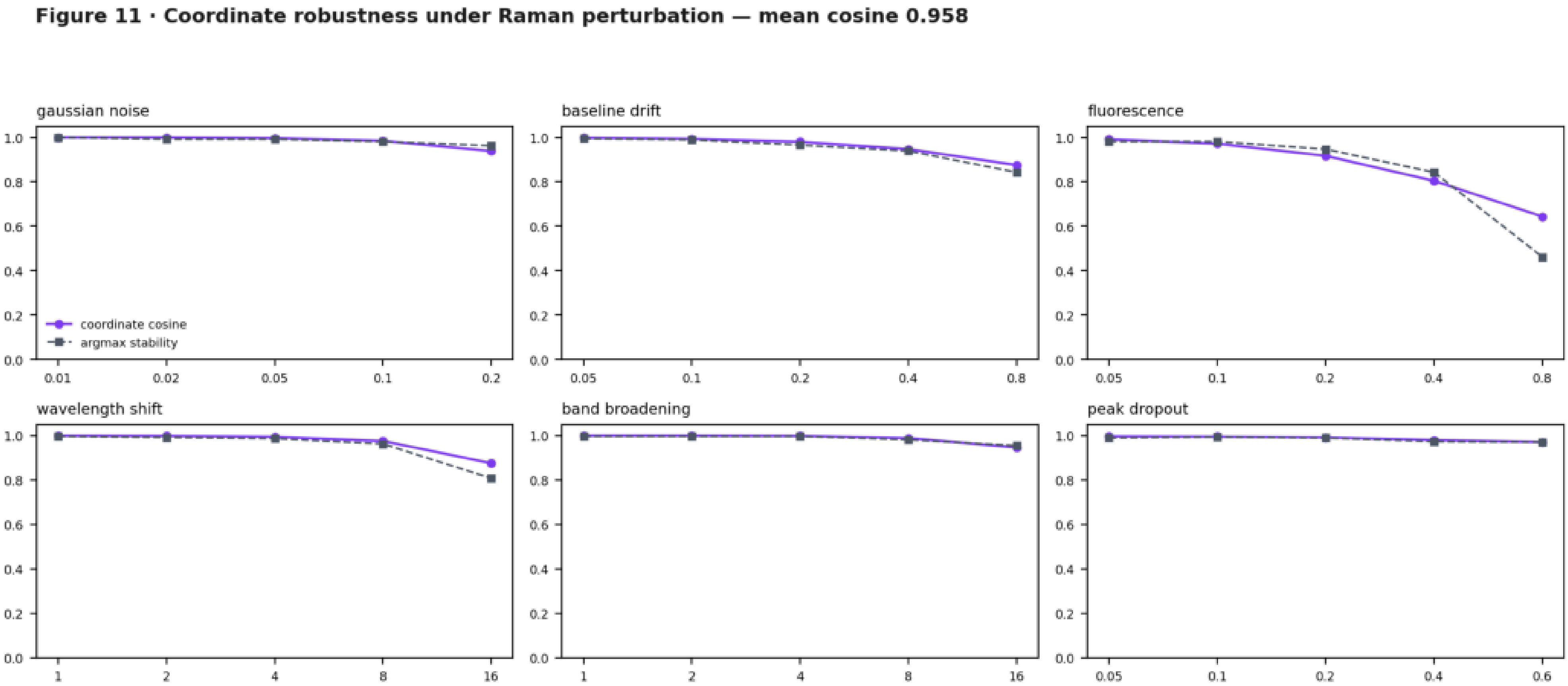
Dendrogram, distance distribution, split gains and the shape verdict. The space is modular (z = 40) and tree-like (cophenetic 0.870) while having no preferred cut height - a hierarchy with continuous branch lengths. The two intrinsic-dimension estimators disagree by 3.3x, so neither is quoted.

Figure 10



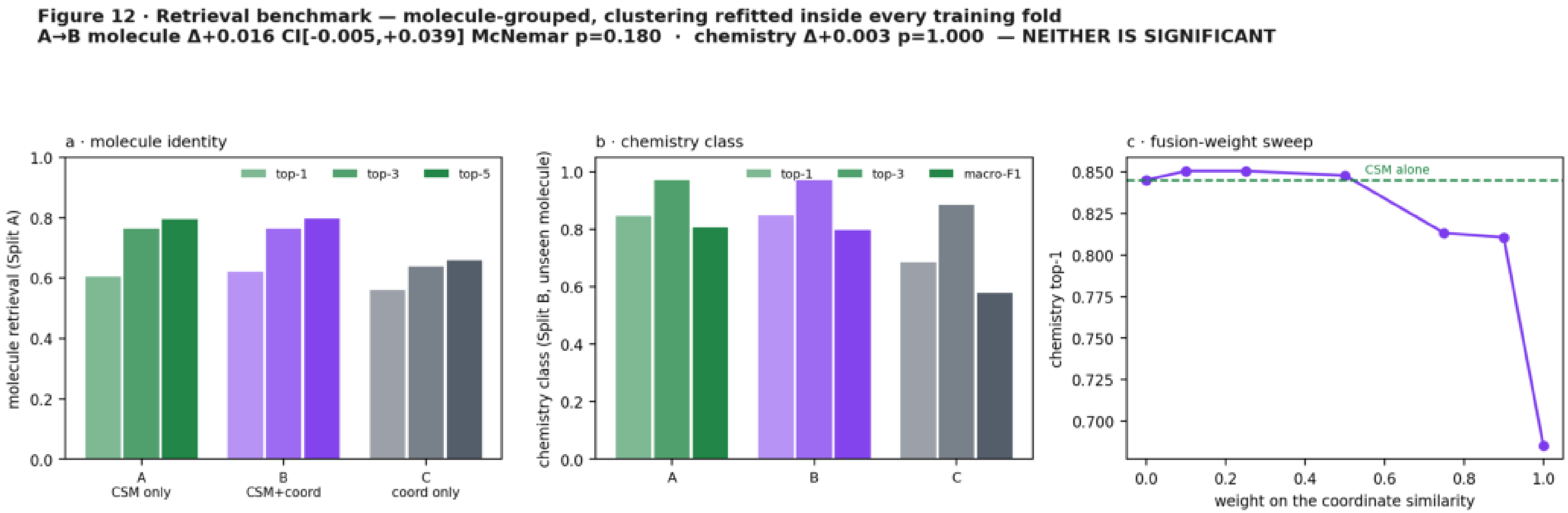
Continuous Spectral Coordinates: kernel x temperature sweep selected on label-free neighbour preservation, and the comparison against hard cluster ids. The coordinates carry nearly double the neighbourhood information of a cluster id.

Figure 11



Coordinate robustness under six Raman perturbations at five levels each. Mean coordinate cosine 0.958, argmax stability 0.949.

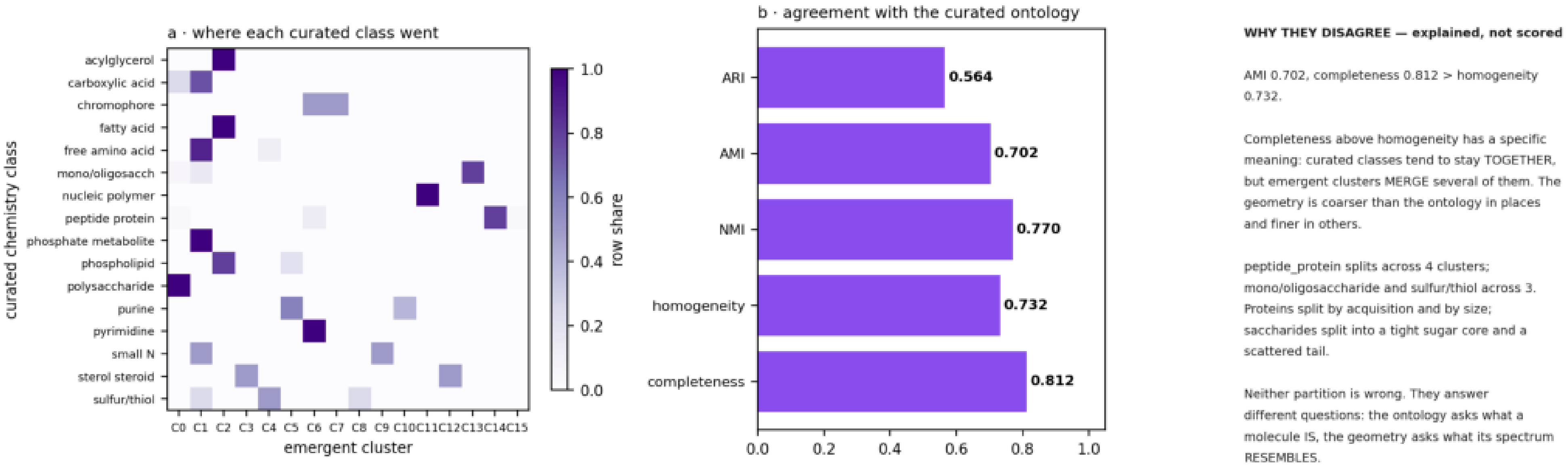
Figure 12



The retrieval benchmark, molecule-grouped with the clustering refitted inside every training fold. Adding the coordinate prior changes molecule top-1 by +0.016 with a confidence interval crossing zero ($p = 0.180$) and chemistry top-1 by +0.003 ($p = 1.000$). NEITHER IS SIGNIFICANT. This is the finding the recommendation rests on.

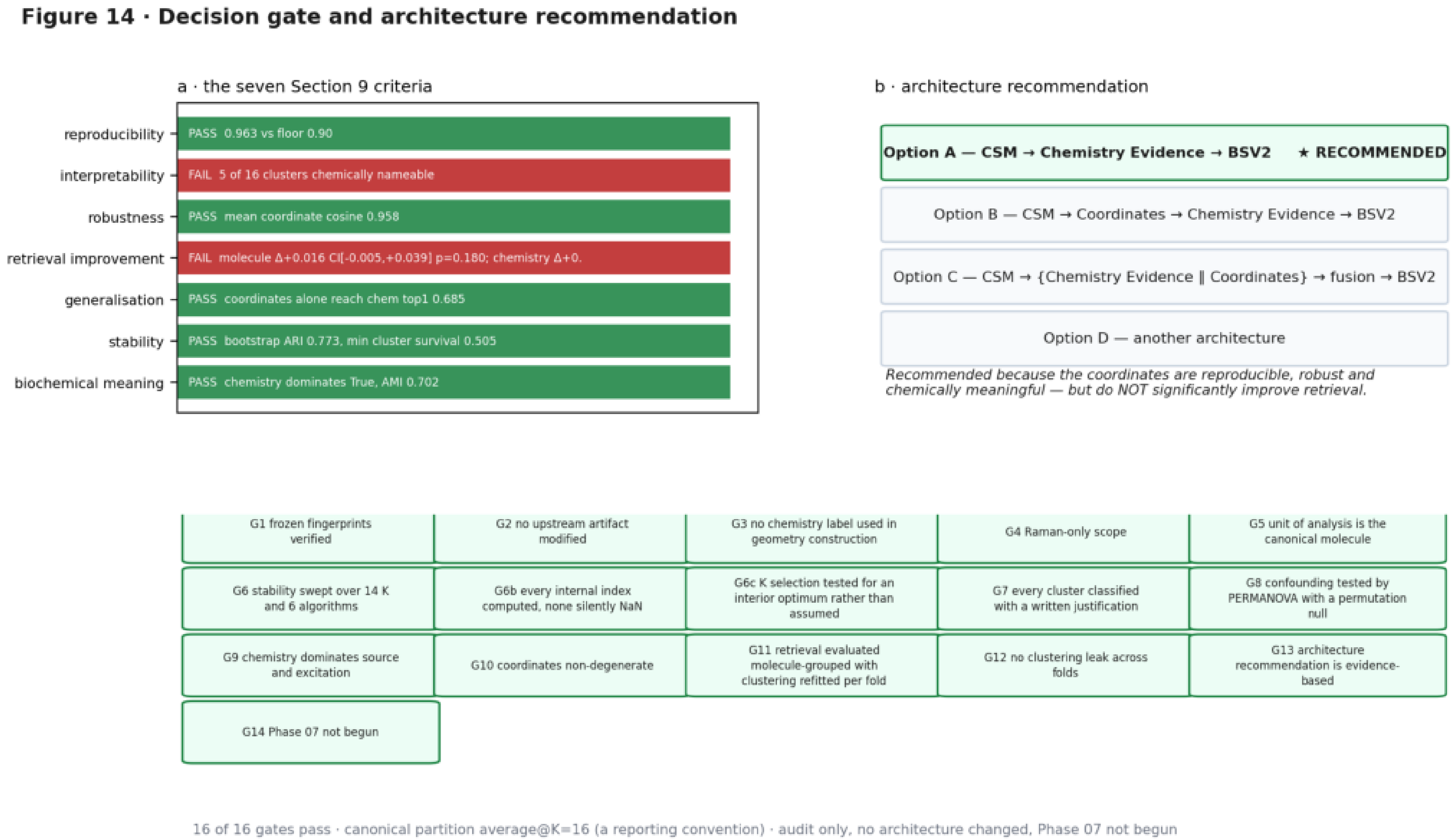
Figure 13

Figure 13 · Emergent geometry versus the curated 16-class ontology



Emergent geometry versus the curated ontology. Completeness (0.812) exceeds homogeneity (0.732): curated classes stay together while emergent clusters merge them. Neither partition is wrong - they answer different questions.

Figure 14



Decision gate and architecture recommendation. Five of seven Section 9 criteria pass; the two that fail are the two that would justify an architecture change. Option A is retained and the coordinates remain a scientific instrument.